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10th & 11th
Short question of
Past Papers 2014-23

{2023}

(3) Inductively coupled plasma mass spectrometry is highly suitable for analyzing heavy metals in the environment due to its high sensitivity and precision.
*Some other techniques also used for this purpose:

- Atomic Absorption spectroscopy
- Inductively coupled plasma Optical Emission spectroscopy
- X-ray fluorescence spectroscopy



(14) IR spectroscopy is used in environmental monitoring for following purposes:-

- Detection of Air Pollutants:

IR spectroscopy can identify and quantify gases such as CO_2 , CH_4 and NO_x in the atmosphere.

- Analysis of soil contaminants:

It helps in detecting organic pollutants, such as pesticides and hydrocarbons in soil samples.



{2022}

(11) AAS is used to analyze a wide range of pollutants, particularly

heavy metals and metalloids, in various environmental samples.

Common pollutants analyzed by AAS include:

- Lead
- Cadmium
- Mercury
- Arsenic
- Chromium
- Copper
- Zinc
- Nickel
- Manganese
- From inorganic book

These metals are commonly found in water, soil, air and industrial waste.

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(9) On Page no. 292

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(10) High-Performance liquid chromatography has several key features:

- Provides precise separation of components in a mixture.
- Fast analysis compared to traditional chromatography method.
- Detects low concentrations of analytes.
- Applicable to a wide range of samples.
- Consistent and repeatable results.

- Advanced systems with automated sampling, data collection and analysis.
- Give both quantitative and qualitative analysis.
- Compatible with multiple detectors.



(12) There are following limitations of GC:

- Suitable only for volatile and thermally stable compounds.
- Ineffective for large biomolecules.
- Extensive preparation needed for non-volatile or thermally labile samples.
- May degrade sensitive compounds.
- Less effective for highly polar compounds without derivatization.
- Expensive equipment and maintenance.
- Frequent calibration and validation required.
- Needs skilled operators for proper handling and interpretation.



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R {2021}
(7), (10)

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{2020}

R (7)

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{2019}

(i) Environmental monitoring involves continuous or periodic assessment of environmental parameters, such as air quality and water quality, to track changes over time. While Environmental analysis refers to specific investigations to identify and quantify pollutants or contaminants in environmental samples using laboratory techniques like chromatography and spectroscopy.

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{2018}

(9) On page no. 311

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{2017}

(3) The principal of IR spectroscopy is based on the absorption of infrared radiation by molecules.

Due to comparatively low energy associated with IR radiation, they are not capable of causing electronic transitions in a molecule. Therefore they cause vibrational and rotational energy changes in a molecule which are also quantized. When the radiation interacts with sample, certain wavelengths of light are absorbed by the sample's chemical bonds. The amount of absorption at each wavelength is characteristic of the types of bonds present in the sample. By measuring these absorption patterns, IR spectroscopy can identify functional groups and provide structural information about organic and inorganic compounds.

R $\longleftrightarrow$ {2016}
(7), (8)

R $\longleftrightarrow$ {2015}
(8)

(7) On page no. 286



{2014}

R

(7)(8)

